Anish Karak

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Education

Kalinga Institute of Industrial Technology — Bhubaneswar, Odisha, India
B.Tech in Computer Science and Engineering; Current CGPA: 9.24 — *Aug. 2024 – May 2028*

Purv International School — Durgapur, WB, India
CBSE (Class XII) — *Apr. 2022 – May 2024*

St. Xavier's School — Durgapur, WB, India
ICSE (Class X); Grade: 95% — *Apr. 2011 – May 2022*

Technical Skills

Languages: Python, C++, TypeScript, SQL, JavaScript, C, HTML/CSS
Frameworks/Libraries: FastAPI, OpenCV, YOLOv8, ONNX Runtime, XGBoost, Scikit-Learn, NumPy, Pandas, React, Next.js, Tailwind CSS
Databases & Cloud: PostgreSQL, SQLite, SQLAlchemy, Firebase Firestore, Cloudinary
Tools & Platforms: Git, GitHub, Linux/Unix, Docker, Postman, Vite, VS Code
Core Concepts: Data Structures & Algorithms, Machine Learning, Entity Resolution, Digital Image Processing, OOP, DBMS, OS, Computer Networks

Projects

Amazon ML Challenge – Multilingual Entity Resolution | *Python, XGBoost, Scikit-Learn, Pandas* Sep. 2026
- Engineered a large-scale entity resolution pipeline resolving **1.73M** noisy commercial records across 336.9M candidate pairs, designing an 8-rule inverted-index blocking engine achieving a **>99.99%** search space reduction.
- Engineered 46 dense pairwise similarity features (lexical, phonetic, address token overlap) and trained a 1,000-tree XGBoost classifier with dynamic relative-gap pruning, achieving **0.805 Macro** $F_{0.5}$ on validation.
- Secured a **Top 500** placement out of thousands of teams in the 72-hour Amazon ML Challenge on Unstop, maintaining zero data leakage via isolated entity-cluster validation partitioning.

Real-Time Object Tracking & DIP Inspection Suite | *Python, OpenCV, YOLOv8, NumPy* Aug. 2026
- Architected a modular tracking system benchmarking classical filters (CSRT, KCF, MOSSE) against YOLOv8n with BoT-SORT and ByteTrack, featuring target ROI selection, trajectory telemetry, and template-matching recovery.
- Engineered a freeze-frame diagnostic canvas executing 8-stage image processing (CLAHE, 256-bin histograms, spatial gradients) on target ROIs; evaluated CSRT across 200 benchmark frames with **100% tracking success** at **35.88 mean FPS**.

KarigarSaathi – AI-Driven Artisan Market Linkage | *FastAPI, ONNX Runtime, OpenCV, React* Aug. 2026
- Engineered the core computer-vision microservice in FastAPI using ONNX-quantized `u2netp` and OpenCV CLAHE, providing automated background segmentation and shadow normalization under a strict 512MB RAM constraint.
- Integrated Sarvam AI Indic speech-to-text models with prompt-grounded fact extraction, transforming regional voice inputs across 5 Indic languages into structured bilingual catalog attributes with zero hallucination.
- Collaborated with Team SYNAPSE for **Smart India Hackathon 2026**, connecting the vision pipeline to an offline-first React/TypeScript PWA with tamper-proof QR Craft Passports and ONDC export schemas.

Wasabi (LabAI Hackathon) | *Next.js, FastAPI, Python, Groq (Llama 3), Bright Data* June 2026
- Collaborated to build an autonomous competitive intelligence platform that scrapes real-time web market signals via Bright Data and synthesizes executive briefs using Llama 3 on Groq LPUs.
- Designed an in-memory caching and state synchronization store in FastAPI, decoupling real-time UI dashboards from web scraping latency and ensuring predictable sub-second endpoint throughput.

Achievements & Extracurriculars

Competitive Programming & DSA: Among Top 12 finalists in CodeSprint 3.0 (USC.KIIT) coding competition with 100% consistency; active LeetCode solver.
Hackathons & Innovation: Selected for Smart India Hackathon (SIH) 2026; LabAI Hackathon participant.
Machine Learning: Kaggle Rogii Wellbore Geology Prediction competitor; Top 500 in Amazon ML Challenge 2026.
Strategic Pursuits & Athletics: Competitive chess player (1500 rating on Chess.com); regular endurance distance runner.